

ActInf GuestStream 059.1 ~ David Tuckett, "Conviction Narrative Theory"

GuestStream | David Tuckett | 1:17:49

Hello and welcome everybody.

It is October 18th, 2023, and we're here in Active Inference Guest Stream 59.1 with David Tuckett on Conviction Narrative Theory.

So David, thank you for joining very much.

Looking forward to this presentation and discussion.

So to you.

So thank you very much.

Just perhaps a bit of a useful background is that I'm someone who started out as an economist in the days when economists were taught more than they are now.

That is to say also sociology, even a bit of politics.

And then I became a medical sociologist while at the same time becoming a psychoanalyst, which I won't attempt to explain, but is obviously different.

And then in more recent years, I've tried to put those things together with going back to economics and trying to understand particularly financial markets, but more generally decision making and how it works are sort of on the big scale.

So conviction narrative theory is the outcome of this kind of exercise.

I suppose it's it most immediately started from the fact that I began a series of interviews, but perhaps I'll just come to that in a minute.

So what is conviction narrative theory?

The purpose of it is that it tries to characterize the social and informational context in which decision making occurs.

And secondly, the cognitive and affective processes governing it.

So it's about how people actually take decisions, not how they ought to.

So this book here, *Minding the Markets*, was a book I published in 2011 and was the outcome of 52 interviews with fund managers.

Those are people who worked for some of the large firms like Goldman Sachs or others and were responsible for investing at least \$500,000 into the world stock market.

Some of them invested as much as 20 billion.

So they range.

And when I was talking to them, I became aware that they were very intelligent people.

They often had up to 20 people supporting them or also very intelligent and able.

They had lots and lots of computer power and programs and analytic schemes and so forth.

But when it came down to it, what they had to do was to make a judgment and they had to be convinced about that judgment.

And not only that, they had to hold to that judgment over time, which is something that's not often discussed in decision making.

So if you decide to sell or buy or even hold any kind of investment, that's a decision you make at point one.

But usually it's not going to pay off to you or the whole idea is it won't pay off for three, five, ten years.

And so you've got to be willing to stick with your decision for quite a long period of time.

Or if you decide you've got it wrong to back out.

And all of this requires what I realized requires what I call conviction.

So that's the basis of where this theory started from.

So real world decision making, which is what we're trying to understand here, is best understood by example.

For example, the one I've just given, how to manage an investment portfolio, but also a question like what level of reserves does a bank need to be safe?

A decision like should you expand your company into a new product or new technology?

Or how much funding should a university or a company allocate to cybersecurity threats?

What resilience standards should be adopted by government regulation?

Precisely what should we prepare to do to prevent catastrophic climate change?

How should we anticipate and prevent future potential financial crisis?

Or how do you level up regions of your country or the world which are currently unequal?

These are all pretty big, big questions.

And I would argue all of them involve uncertainty.

So they're massively consequential choices.

That's the first point.

The data is always going to be incomplete.

The options are ambiguous.

And the future, that is what you know has happened so far, not only may not resemble the past, but in our modern world, which is changing rapidly, is actually unlikely to resemble the past.

Or you could put it another way, which bits of the past will it resemble and which it won't?

For all these reasons, the axioms of standard decision theory are not satisfied in this situation.

Now, the choices that require, to make choice in this situation require both commitment and monitoring, so as to adapt smartly over time as events unfold.

So in the investment example I began with, you have to be willing not just to make the decision, but to stay with it.

It's a bit like a marriage or any other kind of long term choice.

Of course, you can get out of it.

But the point of doing it is to try and stay with it.

And then it will also involve monitoring.

That is, trying to assess if in fact you've made the wrong decision, so you adapt pretty quickly over time as events unfold.

Now, the context of such choices can be labelled as radical uncertain, radically uncertain, because data may or may not be useful and could even mislead.

Now, we should be able to say just uncertain, or even we should be able to say risky.

But the problem is these words, uncertain, uncertainty, risky, have all become given very specific meanings in

different disciplines, which makes it extremely confusing.

But the underlying point here is that I'm talking about a situation where the future is yet to be made, and so it must be imagined, however you're going to do that.

Okay.

Okay.

Now, conviction narrative theory, which tries to address this situation, is designed as an alternative, cross-disciplinary theory of decision making that starts from the proposition that human cognitive, affective and social capacities are well adapted for uncertain contexts.

Now, a great deal of the existing research, at least in psychology, has tended to demonstrate that humans are not very good at decision making, do all kinds of experiments, which show that they have biases, they frame things incorrectly, they don't use statistics properly, they get emotional, etc.

But by the time you finish reading all that research, you wonder, well, how do we get where we are?

You might be a bit, say, yeah, well, look where we are.

But if you look at the advances that have been made by humankind, clearly, we're rather good at making new decisions and at developing new things.

And there must be something wrong with a theory which fundamentally talks about bias.

And one of the ways of thinking about that is to think that perhaps we're adapted to the real world better than we are adapted to the kind of world we set up in labs or the kind of gambling decisions that typically decision making theory tries to start with.

So in both labs and gambling decisions, there is a right answer, or at least there's an answer that consensually people would tend to say was the best choice to make.

So what conviction narrative theory does, and I'll come to it in a moment, is it turns the core issue from how to make best decisions, which is typically what research has been looking up to now, to how to make any decisions at all, given the need to act when in fact there's a possibility of facing loss or failure, real loss or real failure, real disasters.

So how do people do it when they can't, so to speak, or feel there's a really good chance of things working?

And the answer is that they adopt a conviction narrative, and I'll try to explain that.

But the basic idea is they adopt, of course, a course of action, which they believe to be supported by a narrative, which convinces them that by adopting that course of action, they will get,

to the outcome they want to be supported by a narrative, and they will get to the outcome they want.

Now, on the whole, people don't invent conviction narratives just out of thin air.

What they actually do is they adapt narratives, that is, ideas about what actions are effective for the particular thing they're trying to do, from those already circulating in their networks.

Could be in their company, in their social group, etc.

Now, one of the key things that conviction narrative theory is designed to direct you towards is that if you accept that the basic way people make decisions is by trying to adopt a conviction narrative because of the need to act,

then what becomes interesting is how do they do this?

And in order to begin thinking about that, the theory postulates two particular states of people or of organizations, which I'll come back to, in which this conviction is generated.

One is called divided and the other integrated.

I'll try and explain in more detail.

But integrated basically means where people bring all the information available to bear and they're as sensible as they can about it.

They realize what they're doing.

They know it's uncertain, but they're still able to make major commitments.

So it's kind of the academic nirvana, if I should make decision making.

Divided is where, for one reason or another, people only consider part of the story, part of the outcomes, etc.

And what conviction narrative theory is designed to do is to kind of alter the way we approach these problems and so identify different kinds of ways for doing research.

Okay.

Okay.

Now, a central idea that follows from all this is the idea of ambivalence.

And this is an idea that I do get from psychoanalysis, so it's also used otherwise.

And ambivalence is an intrinsic property of any human relationship.

If you consider a marriage, an employment contract, or just about anything which lasts for any length of time, you will tend to have, at any given time, feelings in favour of doing it and feelings against doing it.

It's pretty normal.

That is, feelings generate approach or avoidance.

But, of course, there'll be an overall level at any one time of one or the other.

Once you get avoidance, you get out of it.

Once you have approached, then you basically go on doing it.

And this is based on work going back to Freud or the sociologist Neil Smelser.

Now, decision making in radical uncertainty necessarily evokes, causes people potentially to be ambivalent.

Because while time is unfolding once you've made the decision, the outcome of your actions continues to be uncertain.

And ex ante, you quite simply cannot know the answer.

And that inevitably creates anxiety and anxiety creates avoidance.

There's always a tendency to want to stop what you're doing, perhaps.

Or, if it's the other way around and you've decided not to do something, the argument that maybe you should have can come back.

So, consequently, contemplating action to exploit a perceived opportunity must stimulate a greater feeling of approach than avoidance.

Pretty simplistic theory.

But in other words, approach feelings generate lots of let's do it kind of feelings.

And avoidance feelings trigger loss aversion, inhibition and withdrawal.

Or a defence against that, i.e. just being blind.

Conviction narrative theory, C and T, focus attention, really, on how this ambivalent situation, which I want to stress is both affective and cognitive, is dealt with.

And as I mentioned before, the divided state or the integrated state are two ways of thinking about it. You could think that if you accept radical uncertainty, then most decisions really are experiments. And if you play with that idea, it's quite interesting because making decisions in, so to speak, experimental mode

is rather different than making decisions in other modes.

Because if you're doing it in experimental modes, it means that from the start, you would perhaps build into the situation, as you do with an experiment, ways of measuring what's going on, giving you feedback, formalising your hypotheses as to why you think things are going to work out, and then getting some sort of feedback as to what's actually happening.

Now, obviously, one trick here, the trick that's most disastrous potentially, is you simply eviscerate doubt.

So that, and one of the problems is that a lot of modelling decision, a lot of modelling that is called for in public decision making, is often used, whatever the modelers might have intended, to actually get rid of doubt.

Oh, well, my model tells me it's all right to do it, sort of thing.

Okay.

So to say a little bit more, I've used the term narrative, and so it's worth trying to say what is actually meant by that.

In our theory, narratives are structured, higher order mental representations, which incorporate all of causal, temporal, analogical, and valence information about agents and events, which serve to explain data, imagine and evaluate possible futures, and motivate action.

So there are quite a lot of things that look like narratives, which aren't narratives, unless they have the potential to do all those things that I've just read out.

It may not always be in the narrative, but it is there potentially for it to be a narrative.

So, you know, you would tell a story about what would be the effect of giving this talk, for example.

Now, narrative fragments are subsets of the elements in a narrative which can be readily communicated.

So it doesn't seem to us, after a lot of thinking about it and study, that there's very much to be gained from trying to decide when is it a big narrative, when it's a small narrative, when it's a narrative, when it's a story.

You have to recognise that narratives tend to be made up of other narratives, of smaller fragments on which the main narrative depends.

The second important concept is shared narratives.

What I'm getting at here is that there are narratives that are shared within social networks, which are, so to speak, dominant.

So in a particular group, this is the way we do things.

This is the way we understand things.

Science has lots of shared narratives which govern scientific methodology, for example.

A third important term is explanatory models.

This is taken from social anthropology.

And explanatory models are culturally valid systems of understanding, often expressed as narratives.

So, for example, a theory of economic growth is explaining how economic growth takes place.

A theory of COVID infection and so on.

Finally, a crucial concept is feelings.

And this is based on the idea that feelings, and modern neuroscience is demonstrating this pretty strongly, that feelings are at the centre of all brain activity.

That is, most things you can observe when you look at brain scans and things like that, you typically see movements between different parts of the brain and often going through parts of the lower brain, et cetera, the amygdala and areas of that kind.

The function of feelings, it seems, is that they force their way to consciousness in the human brain, and they prioritise conflict resolution in context.

And this is particularly important.

And it's especially important for economists who, for example, have a theory of utility.

Because I think what we learn from what we know about feelings in the brain is that your utility function alters according to context.

And the brain is sort of set up to do this.

So we actually have a number of conflicting aims, which where one or other is turned on according to the particular context.

For example, to look after people caring, as opposed to enjoy ourselves and so on and so forth.

The overall effect is we feel we want to approach or avoid, including when you're listening to me.

Some of what I'm saying may be sort of switching you off completely, where you may be struggling to listen or even not want to listen at all.

Other bits may be making you feel yes, yes.

And this is going along with whatever cognitive judgments you're making as well.

So to summarise, I don't see, although it's useful to talk about cognition and feelings maybe separately, they always operate together.

That's a picture of the brain to show the point I made, which I won't develop now.

But the fundamental point is that for a long time in the 19th century, the brain was understood the wrong way around.

That is to say, just because the cortex is on the top there and information comes through the cortex.

Actually, it's the lower parts of the brain which are fundamental to conscious decision-making.

And the cortical function is kind of added on top.

Now, there are four functions of conviction narrative theories which explain why conviction narratives work as they do.

This slide doesn't seem to work.

So the first function of a narrative, in conviction narrative, is explanation.

That the narrative imposes structure on the current situation that yields a sense of understanding and emotional satisfaction.

So, for instance, typically when people do science, they do all the stuff they do with the methodology and so forth and they present results.

But in doing so, they tell a story about it, explaining the situation. And that we do.

Most stories have an explanatory quality.

The second element in a conviction narrative is it allows a mental simulation.

So, for example, if you are thinking about, shall I adopt this particular way of trying to do climate mitigation, I can think of the actions,

I can imagine,

I can get all sorts of spreadsheets

and proper hard information,

and I can try and look at what I think

will be the outcome of that particular activity

in 20 years' time.

And as I do it, in some way or other,
I'll be forming, at least implicitly,
pictures in my mind of what the earth looks like
when I finish doing all this.

And then you can compare an alternative
or many alternative narratives.

And so to speak, you run the narrative forward
to generate imagined futures
associated with a particular choice.

The third, which follows from what I said about feelings,
is that these explanations and mental simulations
are evaluated affectively.

So you react emotionally to the futures you imagine
to evaluate their desirability,
and that helps to manage commitment over time.

And the fourth and really important function
of narrative in a conviction narrative
is communication.

Most important big decisions of the kind
I'm interested in involve a number of people
working together.

It could be several labs, for instance,
or it could be people working to a particular aim
in regard to climate or many other problems.

And so when you work together with other people,
you have to motivate them
and you have to explain to them
and you have to give them an idea
of where you're going and what the point is.

And we use narratives for that
and narratives that propagate across social networks
and become in that way maybe shared narratives.

So the other thing I'd like to stress
is the conviction narrative is not any old narrative.

It's a narrative that sets out
how you're going to change the current situation
by an action to lead to a desired outcome.

Okay, it's very specific.

And to do that, you draw on narratives
in your context.

So the narratives solve two particular,
or there are two sets of problems
in decision making if you think about it.

We call this the logic of decision.

So that decisions reflect both data picked up
from the external world,
including the social environment,
and internally derived goals.

So to make a decision,
you've got to do both things.

You've got to pick up data
and then add it to your goals.

The first aspect of this,
you can call the mediation problem,
which in the diagram is the dashed lines.

It reflects the need for an internal representation,
what in our paper we call a currency of thought,
that can mediate between data from the external world
and actions decided internally.

And one of the things about economics,
for those who are economists,
is that economics actually has,
as far as I can tell,
no theory of cognition.

Models in economics,
you just write down,
you know, R for real rates.

It doesn't have a theory
of how people come to know
what the real rates are.

And of course, as soon as you do have
to fix such a theory,
you've got lots of room
for disagreement and differences.

Second is the combination problem,

which is the grey lines.

And that reflects the need for a process,
a driver of action,
something to get you actually doing something
that combines beliefs and goals
to yield actions.

And in classical theory,
classical decision theory,
the currency of thought is probability.

And the driver of action
is expected utility maximisation.

So, you know,
if you've got a probability
of 70% of success
and that's good enough,
then what you're supposed to do
is to choose the highest thing.

And that is a very seductive
and sensible way
to make decisions,
provided you think
you've got probabilities.

In behavioural economics,
the main emphasis is on
how people get the wrong probabilities.

They make decisions based on biases
and so on,
but they don't take the probabilities
properly into account.

In radical uncertainty,
the argument is
there aren't probabilities,
or at least there aren't probabilities
for the kinds of questions
that I've been trying to talk about.

So in CNT,
the currency of thought
we propose is narratives

and the driver of action
is affective evaluation
of narratives in a social context.
And this is a rather complicated slide
to say something about that.
You see that, you know,
we have data,
we have the narratives,
we have this explanation and simulation,
there's affective evaluation
leading either to approach
or avoidance,
link with your goals,
and then you take action.
So that representation
and processes
in conviction narrative theory,
narratives are supplied in part
by the social environment.
It's a very important part
of the theory
that we're picking up narratives
from around us all the time
are used to explain data.
They can then be run forward in time
to simulate imagined future outcomes
of action,
which are then evaluated effectively
considering the decision maker's goals.
And these appraisals of narratives
then govern our choice to approach
or to avoid these imagined futures.
And you see from the figure
there's lots of feedback loops.
Okay, and I'll just talk a little bit
about divided and integrated states.
So as I mentioned,
this diagram is supposed

to illustrate narrative prediction,
that is what your narrative
is going to tell you to do
as you get new information
based on where you started your prior.

So insofar as the new information
you get creates congruent feelings
to where you were before,
you can pay attention to that data.

In an integrated state,
you pay attention
whether it's congruent
or not congruent
with your existing narrative.

In a group feel
or divided state,
how it works is
you only pay attention
to the things
that you already agree with.

In other words,
confirmation bias
and incongruent
information
that produces
incongruent feelings
is ignored.

So this is how you get
bubbles in finance
and bubbles in lots of areas
of activity in society.

Because if you don't
if you don't attend
to the negative information,
then that's the only way
you can go.

Bear in mind,
this can also work the other way.

If, for example,
you're in a depressed
or in a state
where we're not doing anything,
you can continue to
there you discount
the positive information.

And so you stay
in a repetitive bubble.
So an integrated state
group of affairs
is that envisaged
in normative theories
of science
based on experimentation
and or inference.

Strategic actors
are curious.

A broad range
of information
and perspectives
both supporting
and challenging plans
and actions
are considered.

And it's recognized
that there are uncertainties
and contradictions
within the preferred
conviction narrative
supporting a strategy
or policy.

And crucial
to the integrated state
is the idea
you know
that although
you're making a decision

it could be wrong.
So you're open
to new information.
The divided state,
I won't go through it again,
is just the opposite of that.
Okay,
this is just
a last slide
to just give
a quick link
to work that's been done
particularly in psychology
where you can see
that this whole theory
is very dependent
on what's going on
in your local
social environment.
Because in your local
social environment
there are various
narrative elements
influencing your evaluation
of narratives.
narratives.
So,
you know,
the stories
that are around,
the shared history
of the group,
various heuristics
or logics
or frames
or conventions,
beliefs and so on
which are all work.

But these interact
with the three things
which have been found
very significant
in lots of psychology,
what are called
content effects,
that is ways
in which different content
tends to be picked up
positively or not.

Trustworthiness effects
that there are all
sorts of effects
about listening,
you know,
you can dress somebody
up to look like
a professor
and people
are supposed
to believe them
more as opposed
to a tramp
or something.

And presentational effects
which has to do
with the way
things are presented
and so on.

This can influence
whether a narrative
is convincing or not.

And overall,
you get the usual
thing that it all
leads to a narrative
which either generates

approach or avoidance
and only if there's
a positive balance
do you go for action.

Okay.

So that's roughly
the theory.

I hope it makes
some sense
and generates
some questions.

All right.

Awesome.

Okay.

Well,
while I am
cropping
and returning
everything to normalcy
visually,
how did you
come to this?

I mean,
what brought you
through the warps
and the weaves
to want to highlight
this perspective?

Well,
so first of all,
I think when I was
lucky enough
to study
economics
at King's College,
Cambridge
it would have been
the college

of John Maynard Keynes.

economics

and when I was

there,

it was still

very much,

we were taught

by Keynes pupils

and they saw

economics as very

much to do

with the management

of uncertainty.

Economics

today

has defined

uncertainty

in such a way

that it doesn't

have to take it

into account

and so everything

can have a probability

and so on.

So I've always

felt uncomfortable

with what's

happened in economics

because the real

world seems to me

uncertain,

as I said earlier.

You know,

there's the whole

point about human

innovation

and the way

in which new

companies are formed
in an economy
or new scientific
theories come about,
et cetera,
is they're new.

They may be
recombinations
of the past,
but they are
new and it's
not helpful
to have a kind
of equilibrium
theory for those
sorts of problems.

However useful
it might be
for other things.

And so that was
a general background.

It was through
psychoanalysis
that I came to see
the importance
of feelings.

Most people,
when they think
about psychoanalysis,
probably haven't,
don't know much
about it or
had a rather
half-baked
explanation given.

But actually
what Freud's
major contribution

is,
was that he
understood
that,
so to speak,
feelings and
thoughts are
intricately linked.

That if the
thoughts you're
having make you
feel bad,
you'll do all
kinds of things
not to have them
or even not to
need to have
them in the first
place.

So this link
between thoughts
and feelings
has always been
around.

It actually came
in part from
his neuroscience
but in more
recent times
it's now
become,

I think,
a largely
accepted view.

I gave this
theory,
the outlines
of this

theory to,
I tried out
on Carl
Friston,
who probably
most of you
know,
about 10 or
11 years.
And Carl
is a great
listener.
He sat there
and he listened
for about 15
minutes and he
looked very
thoughtful and
then he said,
well,
there's nothing
you've said
which is in
opposition to
current
neuroscience.
And so that
was nice.
That encouraged
me to go on.
Anyway,
so also,
as I've
mentioned,
when I began
trying to
understand how
could the

dot-com
crisis in
financial markets
have come
about,
which I was
studying just as
the global
financial crisis
was actually
not happening
but about to
happen,
it was then
that I
realised,
and I think
quite a lot of
people have
realised it
afterwards,
that this
kind of
thing in
which people
make decisions
based on
convictions,
they do the
best they can
with the
information they've
got,
but by its
nature,
what they do
is uncertain
and therefore

fragile.

Awesome.

Yeah,

definitely,

while there

wasn't any

active inference

generative models

in the talk,

you brought up

so many

motifs that I

think we can

explore a little

bit because

they're the

kinds of

things that,

as you

pointed towards,

necessitate

a cognitive

account.

So,

first,

how would you

see cognitive

modelling

playing a role

in micro

economics or

macroeconomics?

What kind of

cognitive model,

what kind of

explanation,

prediction,

or intervenability

do you
seek?

Well,

I mean,

there's no doubt

that

as we

go about

our business,

we form

cognitive

models of

the world

around us

models,

and

in my

opinion,

that's a

useful way

to think.

And I

really call

them explanatory

models because

I first

encountered them

in anthropology.

This was when

I did a big

study in the

mid-80s of

a thousand

patients going

to their

doctors and

looking at what

they'd understood.

understood.

And to cut a

long story

short,

although at

the time it

was widely

thought that

patients didn't

understand most

of what their

doctors said,

and therefore it

wasn't really

worth their

doctors spending

much time

talking to

them,

in fact,

what we

found was that

most patients

did understand

what their

doctors were

saying,

but where

they didn't,

it was because

the model of

understanding what

was wrong with

them that the

patient had,

so they had

their own

model,

was not the
same as the
model the
doctor had.
And they
hadn't established
the two
explanatory
models and
then put them
together.
And in fact,
very often it
would have been
quite easy for
the doctor if
he knew what
the patient was
thinking to
point out that
this, that,
or the other
assumption or
process, as we
would say,
you know,
wasn't how
things worked.
And then that
would have helped
the patient to
understand.
As it was,
the patient just
went away not
quite understanding
what had gone
on and often

didn't follow
the treatment.
So that's how
I got into,
if you like,
cognitive models.
Now, when you
try to formalize
them, you know,
I think it's
extremely interesting
to try and
formalize them.
But I always
have the
reservation that,
you know,
where, it's
always where
are we getting
the information
from and how
robust is it
to changing
circumstances.
So I mean,
that's the,
that's I think
the key problem.
cool.
So I was very
interested by
that provocative
example of the
person making
financial decisions
with, with a
20 person team.

That's a lot of attention and energy, a lot of skin in the game just for that kind of an organization to, to deploy that kind of a decision support setting.

So how do we think about that?

What, what works or doesn't when there's group decision making?

And then how does that integrated, divided dialectic within an individual relate to the status of the integrated or divided understanding that might arise at a group level?

especially if it's a decision where like, it's obvious from the get-go that no, no one of those people know whether the stock is going

to go up or
down.

So then what
does that 20th
person or the
21st person or
the 22nd person
do on that team
to actually
empirically lead
to better
outcomes?

That's a really,
it's a really
good question.
question.

So in order
to, to, to
think around it,
there's certain
background facts
that are important.

One is that
these people doing
that work are
on the whole,
the most highly
paid people in
our society,
which is, which
adds a level of
fascination to
your question,
because we know,
we know from
academic research
that no one,
no, there's no

published data
anywhere to
suggest that
anyone can
consistently do
better than the
average.

Okay.

And the reasons
are obvious.

Nonetheless,
there's always
hope, right?

And so one of
the ways you
could start to
think about it
is, is this
supposing it's,
it's your
pension,
supposing it's
your pension
and you've got
a choice,
to put the
money in
something which
will just get
you the average,
which of course
will mean that
at times it'll
go down,
at times it'll
go up,
you'll just get
the average,

versus somebody
who says,
well, you know,
I can do better
than the average,
look at my last
year's results,
and last year's
results will show
better than the
average, because
obviously by
statistics,
some people do
better than the
average.

So then it's
quite difficult
for people,
it seems,
to say no to
the possibility
of doing better.

And that might
be one part of
the story here,
but at the
level of the
people with
20 teams,
they essentially
are paid to
do this work,
work.

We know that
they do it
better than
you or I

could do it,
on average.

So there are
studies that
show that
individual
investors don't
do as well as
professional
investors.

But there are
increasingly studies
that active
investors,
that is people
doing sort of
what I did,
you know,
who look over
the data and
try and decide
what to do,
that active
investors in
fund management
don't, on
average, beat
some form of
more computer
based, automated
way of doing it
which just looks at
averages and so on.

Of course, all of
these methods fall
apart under certain
situations.

Computer based

methods did
terribly badly
at certain
times and they
have done
recently.
and we also
know that if
you, so that
if you know
what private
equity, if you've
heard of private
equity or venture
capital, this is
another level which
ordinary people
aren't allowed to
invest in because
it's supposed to be
risky, but where
actually they often
do do better
though because
they've got less
money to play
with and they
can use expert
knowledge in
particular areas.
Again, it's
subject to all the
problems I've
talked about.
So why, what is
all this about and
why are all these
people engaged in it?

I think the
answer is that
there's a constant
attempt to
support the
narrative that,
you know, what do
you do with your
money?

So, I mean, the
university's got,
the university
pension fund's got to
invest, right, your
pension.

It's got to be done
somehow.

And, yeah, I
could go on a
great deal because

I think the
conclusion from
this is that it
doesn't work very
well and that's
because what
people do is
they pursue what

I call in the
book fantastic
objects.

That is, they
become convinced
that, you know,
the new way of
doing hydrogen
technology or the
new way of doing

this, that or the
other, which is a
way which is
promising but which
nobody knows enough
about to be sure
about.

That's going to be
the thing to make
you your millions and
that's what you'll
invest in.

Or there's ABC
person who's a
fantastic investor.

He becomes a
fantastic object.

Okay.

So that, what a
fantastic object works
with group think in a
divided state to get
rid of doubt and
then create this
process I described
earlier, which is a
sort of self-fulfilling
prophecy process, which
works until it
doesn't.

Cool.

Another piece I found
really interesting is
the laboratory
decision-making
setting, whether it's
reductionist or however
we want to

characterize it, it focuses on certain kinds of decisions. First off, ones that align well with the axioms of standard decision theory, but just pragmatically, they tend to be these one shot gambling decisions or game theory decisions that are really amenable to the kinds of matrix calculations and the Nash game theory and all of this.

And yet, that is not how it feels to be making decisions day to day.

And then you brought in the approach avoid.

So when the gambling is in front of us and we only have the decision making alternatives, it kind of removes this meta on the decision, like whether one even approaches that slot machine or not, which is a much more nebulous decision because everybody starts in a different position.

And then also this consistency, which is really where conviction comes into play with

the marriage or with the investing, which is even a decision at one time point is really more like an agentic event in a broader tapestry or narrative, whereas in the gambling setting, it can lead one to make models that are sort of like decision next, next, next, next.

And it's like a blank slate every single time that a new decision is made.

And then that has limited ability to transfer in to all kinds of other situations that you describe as having that radical uncertainty, like to the root, rather than a superficial uncertainty, like whether in this one narrow decision, something should happen one way or not.

Yeah, exactly.

I think that's exactly right.

And of course, I mean, it's not really to be negative about all the stuff that's been done because it's shown very interesting things.

But I think the main issue is, have you identified your context correctly?

because we kind of know, I mean, to go to sort of Bayesian stuff, as I understand it, although I don't fully understand how it's really done, there's an argument that we're good samplers, that we sample well locally, people do, on the whole.

But of course, the problem comes when you get out of your context.

context.

And so the, I think, the key issue is sort of the transferability of information or knowledge from one context to the other.

Some friends of mine, Lenny Smith and Erica Thompson, have written a paper called *Escape from Model Land*, which is all about how, you know, they're modelers.

They do weather forecasting and climate models and all that.

But the question is, how do you get out of the model into the real world and what challenges do you then face?

I mean, and that's much

more, of course, economic models are much weaker than those kind of models they're talking about.

But it's a context thing and how can you assess whether you've got the right things in your model, you've got the right data, et cetera.

And these are the things where you tend to, I mean, the financial crisis was partly caused by people having the wrong models.

But they'd seen right at the time.

Yeah.

Well, the active sampling, a good sampler is exactly what active inference is about.

Our eye saccades are in the business of doing good sampling, good epistemic value.

And when the context shifts, like if you have a book and you flip it upside down, your eye saccading is not going to be making as good sampling.

And that's why it's hard to read.

Let me ask some questions from the chat.

All right.

Bert asks, there are many similarities with scenario planning in transport policy.

How could policy planning move from cost benefit analyses to the

use of narratives?

Well, that's a very good question.

There's a paper on there, the second paper down that deals with futures and foresight and convictionality theory.

But so the problem with cost benefit analysis is kind of a logically sensible idea in which, you know, you try to work out the costs of a project and you try to work out the benefits and then you work out whether it's worth doing.

The problem is with all the assumptions, as obviously the questioner realises.

And in a formal cost benefit analysis, you have to make up lots and lots of numbers in order to make it formally work.

So I think the question is whether you're using the cost benefit analysis to have a discussion about making a decision and to open up ways of thinking and question assumptions and all that kind of thing, or whether you're using it to come up with an answer which then you aren't responsible for.

And you can say, you know, the model told me this.

I mean, putting it bluntly.

But obviously the major problem with cost benefit analysis in transport is you have to make all kinds of

assumptions about the benefit of,
you know, going half an hour quicker
or of having fewer people in your
in your railway carriage or over what
period of time you're going to do it.

And we know that most major projects
have always overrun costs, have always
been heavily criticized at the time.

The Sydney Opera House is quite a good
example, a completely iconic building
that was a total disaster when people
first played music in it, etc.

And obviously we have things closer to
home in the UK at the moment which
have this problem.

So I think where conviction narrative
theory can help you with that
process is by first of all helping
you to look at, try to dissect your
narrative.

Where are you getting your conviction
from and which bits of the information
you're getting, which narratives are,
you know, giving up, are being
considered compelling, which narratives
are not being discussed, what points of
view, for example, you know, the working
class people through whom the freeway
gap is not being and not being taken
into account, etc.

So I think it can assist with thinking
around decisions and with using formal
methods, which I'm absolutely not against
using formal methodologies, but using
them properly and checking out their work
in the context.

Cool.

All right.

Another question from the live chat.

So Chagor writes, even if on average generating alpha returns are not possible, is there work that identifies selected investors such as Warren Buffett and others that have consistently identified asset mispricing?

Wouldn't identifying investors that are able to consistently spot mispricing support conviction narrative theory, i.e. sophisticated or successful investors might understand how the narrative could change?

Well, so, you know, I haven't done the research on whether or not there are consistently successful investors.

investors, but as I understand the literature and it's pretty consistent, the answer is no, there aren't any consistently successful investors.

And those who are successful eventually aren't.

I mean, I happen to have quite a close friend of mine, George Soros, who is one of the people who did very well, but he also failed.

If you talk to him, he failed.

And of course, once you get like Warren Buffett, it's complicated there because if Warren Buffett does something, you know, other people do it and so on.

But it's of course possible, as all the statisticians on the listening to this will tell you, there's no problem having a few outliers statistically.

It doesn't tell you that it's anything to do with those individual people.

I mean, it might be.

So, conviction narrative theory would explain how people manage to persuade themselves and each other that their active form of investment works.

And it would also help to be used to identify the basis of their claim.

But on the whole, I think you can understand a great deal of what goes on in the financial markets by deploying the concepts of fantastic object that I mentioned, integrated state and divided state.

Because in an integrated state, it's very difficult to do investment.

Very difficult indeed.

For the reason you said in your questions to me, you know, you said, well, obviously, having 20 people in the room isn't necessarily helped.

But you can convince yourself it helps.

And I can tell you, there are some incredibly imaginative people out there who work incredibly hard.

So, one person I interviewed, he had not only got his team of 20, but he'd assigned them different roles to be for and against things, to do this calculation, to play jokers when they wanted to make a particularly strong point.

And he paid them according to the way they behaved and so on.

He worked really hard to try and get it right.

But it didn't end up.

He was convinced.

But it didn't end up over the long term being any better than any other way.

Interesting.

So, even from our ACDIMP post, we've seen some inroads into cognitive economics, where people are drawing from the recent developments over the last decades in cognitive science, the pragmatic turn, extended, embedded, enacted, encultured, affective, all of these different features that are increasingly being understood in the context of human cognition.

As well as non-human cognition, biological or synthetic or otherwise.

So, where does economics go as it also takes this cognitive turn?

How have you seen economics play a role from what you've worked on over the decades?

And then, like, how would it have been different if there would have been this cognitive turn earlier?

Or how will it be different going forward as there's more of an engagement with economics and cognitive science?

So, I think, first of all, microeconomics.

Microeconomics, in its sort of best light, has small models that tell illuminating stories about the relationships between certain, I mean, simplest one is supply and demand. And it tells useful stories.

However, the question is, is that model working in any particular situation you're in?

So, even supply and demand, you know, what if the, I mean, the model is still true, but it may not explain the situation you're trying to deal with.

And it occurs to me while he was speaking, actually, one very interesting example.

So, there is a high-level academic group that was set up to look at radical uncertainty, which has on it many people much more eminent than me on both sides of the Atlantic.

And on one occasion, a very well-known American behavioural economist presented the work on gambling that you alluded to and gave the group sort of, you know, the same test that he'd given to his

Harvard economics class earlier that week, right?

He was completely astonished when this group of people didn't give the answers that he was expecting and he got out of his very intelligent Harvard graduate class.

And the reason was that people said things like, well, if that bet was available, I know the bank wouldn't pay out.

And in other words, they thought round the problem for things like institutional constraints, so that they didn't just think of the problem very narrowly framed, whereas obviously the graduate students were being trained to think narrowly, which can be useful, right?

So, I think where the fundamental issue with economics, since I studied it back in the late 60s, is that it's become entirely, nearly entirely model based.

There are, of course, exceptions, but the sort of high prestige of the discipline is in producing a model.

And if you read the papers, they're often like this.

Let's approach it with a simple model.

We'll have one person, you know, and they're always simple models.

They're always lots and lots of assumptions.

They work very nicely, but it's totally unclear how you apply them to the kind of world that I'm talking about.

Meanwhile, what economics doesn't do very much of is actually go out into the real world and actually look at what happens, what people do.

So, the study I did of fund managers, you would have thought someone, you know, I mean, I'm an amateur, right?

You would have thought someone in financial economics would have done this a long time before, but they haven't.

They write lots of papers.

They do lots of analysis of very large data sets and things like that, but they don't actually look and see, well, how are people doing it?

So, I think one of my hopes would be that it would, it's not that you don't want models, but you need to understand a lot more about the human context in which they're being used.

And in particular, I think, you know, typical discussions of behavioural economics, for example, is, you know, you compare rational and irrational behaviour, where rational means entirely model based, right, which wouldn't necessarily be rational in ordinary terms, and irrational means anything else.

Now, if you can't apply probabilities, the whole model doesn't really start.

I think that this restricts thinking about, well, how does emotion, for example, enter into decision making?

And there are a lot of studies in the management literature of what happened to Nokia.

You remember Nokia were producing mobile phones, and they were producing most of the Nogar phones

on the planet, and then suddenly they weren't.

And this happened pretty quickly.

And some of the arguments about it concern complex emotional issues in the company, where the people who had done very well for Nokia over many years were essentially engineers who built really good kit.

But they didn't understand and hadn't, they were neither listening to the market, so they weren't like the people trying to sell the things, and nor were they software engineers.

So they hadn't yet understood that actually the hardware doesn't matter much.

It's the software, because if you compare many of those old Nokia phones to an Apple, I mean, the Nokias were engineering superior things in many ways, but the Apple, of course, won with all the things it could do.

So I think you can, you know, you need to, there need to be many more studies of this kind, in my opinion.

Yeah, what you said there about there being legacy prestige in presenting model-based, local, decontextualized, variability-reduced analyses reminds me of ant foraging models, where it seems like every single paper can present a simple model of a single context, making many explicit and implicit assumptions, and then the loop is not always closed by going back out there.

So I guess my question is, how do narrative models help us, or what else do we need to supplement or complement potentially very sophisticated cognitive narrative modeling?

Because if you point to that phenomena in the setting of matrix-based game theory modeling, and people think that just by constructing that model, that it's an open and shut case, well now, here's a more shiny and feature-laden type of modeling that seems to cut closer to the real crux of the issue with sense-making on the inbound and action selection on the outbound.

So how do we work so that cognitive modeling or narrative modeling doesn't also fall into that trap where people just present a simplified cognitive model and then continue the same kind of epistemic gestalt in how it's deployed?

So what else is needed beyond the model or what sits beside the model?

Well, so I don't know whether this is really a very good answer, but I've never been more than a part-time academic, which is no doubt a bad thing in some words.

But I think they're really fascinating.

So models can at least claim to give you general truth.

And the problem about general truth is it may not apply to any specific situation.

But many people find it elegant.

I actually find real interest in problem solving and particularly when you've got, I mean, I think interdisciplinary work is fascinating.

It's not fascinating when you all sit around and each person presents, you know, how their discipline does it. That tends to get boring after a while.

But if you've got a problem to solve for someone or, you know, a problem, then you start to really get to grips with what the different people can bring to the table. You know, and I've been very impressed, for example, working with Lenny Smith and Erica Thompson, who are these weather forecasting mathematician modelers. The way they manage to frame and formulate questions is just fantastic. I mean, I think models can help you think very clearly about things, including thinking where the weaknesses in a particular system are. And I think that that's what you could add to the narrative theory is really doing research to try and unpick the details of the narratives people are putting forward and to understand, you know, that diagram I put up is a very complex diagram, but actually digging into in any given case, such as, you know, the decision to abandon in the UK particular railway project or the decision whether or not to, you know, what we should be doing about heating and insulation over the, you know, these kind of big questions. So they're bound to be lots of narratives. And could you actually look at them very carefully and subject them to the kind of rigorous thinking that you get in this cognitive modeling and so on? That I think is the way forward. But I think to do that, you have to work actually together on the same problem. Then you learn from each other. Cool. And definitely a theme that on one hand doesn't seem to be the formal kernel of active inference, yet we see it arising again and again. Like for pragmatism to blossom, there has to be a real problem and a real situation and a real approach. And it has to be iterated. It's not a spectator sport to work. It's not a spectator sport to be on the field. Yeah. But I think that's a very cool topic. Well, where do you go from here? Like what avenues of research with Carl Friston's Via Negativa endorsement? Where do you see this line of work heading? I think it's got quite a lot of, I mean, obviously there's room in economics. There's also room in what was mentioned about futures and foresight science. And what I hope is people will use the framework to do empirical work. I mean, I don't think you can really ever imagine asking the question, is conviction narrative theory true or false? I mean, it seems to me with most high level theories of that sort,

what it usefully is an orientation onto a problem or a set of problems,
which is very different than standard theory.

It doesn't mean standard theory should be replaced by conviction narrative theory,
but it means that the range of questions and research that people might be doing
could be considerably broadened.

I mean, that's not a wonderful answer.

That's basically, I think the other problem is that to do this stuff properly,
you need, you know, I've had in my team two or three psychologists,
a computer scientist, a mathematician, and a social anthropologist,
but I've never had enough resources to have them all at once.

And that's the problem of interdisciplinary research.

You actually need a bigger team.

And then the team has to work together, which is also complicated.

So I would say that's another very important implication,
that it's too easy at the moment to work in a dugout or in a silo.

What you said there about conviction narrative theory,
it can't be simply validated or invalidated by empirical evidence.

It's a lot like active inference.

Active inference isn't waiting to be established or disestablished
based upon a measurement from a real system.

However, you described conviction narrative theory as an orienting tool.

We could even think about that in terms of the OODA loop, like observe, orient.

Okay.

We observe the publication.

We've oriented to it.

We've updated our cognitive model.

And now the decision and the action are where the specifics come into play
because someone could deploy narrative conviction
and then say that this entity has this conviction
and that may be empirically true or false.

And so it's like all of the validation or invalidation is in the realization,
but the concept is something that kind of sits before the realization.

And so when it is properly understood and respected for being theoretical,
then it can be like this fountain of real testable pragmatic hypotheses,
real value being layered over other results and models,
yet not an account or an explanation in any specific setting by itself.

But it's something that lets us jump off of that orienting platform
into all these settings that we've just discussed,

ranging from like the pure financial investing,
which most people are not going to directly involve in,
to the transport policy and climate issues.
So it has a real scope and it's really cool
just to even hear a little bit about how it's developed over the years
and during a time of at UCL and beyond development of the cognitive sciences.
I mean, there's one other obvious application
which may concern all of us is to research policy,
research applications and how research funding is done.
Because I think it can be applied, you know,
because basically, as we all know,
if you're giving out the money,
you're the one giving out the money,
you've got to develop a conviction narrative
that giving money to X is going to do Y
is going to achieve whatever you're trying to achieve, right?
And at the same time, you can see that what X is doing
is trying to develop a conviction narrative to persuade you.
Now, one of the things about many funding councils at the moment
is that they typically talk about doing innovative work, high risk.
You know the various phrases, high risk, innovation, etc.
And then what they basically do is give the money
to people who look reliable with a good track record,
which are probably contradictions in contradictory activity,
which isn't to say you shouldn't give money to the usual people.
But there's clearly a problem.
I mean, for example, on the book I've put my slide up,
Minding the Markets,
you'll see that George Akerlof,
as a Nobel Prize winning economist,
was kind enough to endorse it.
And when I was talking with him,
he said that his Nobel paper had been turned down
by three journals, I think it was.
And when he got to the dinner for the Nobel laureates,
there were three or four others at the table
who'd all had the same experience.
So, of course, there are lots of bad ideas

that don't get funded.

But the issue of how people develop conviction,

which is really how do they manage doubt?

And doubt and the possibility your reputation will,

you know, if you give money to some incompetent group

or waste it all, right,

that your reputation will suffer.

And we know if you go to Whitehall,

if you go to government,

people are terribly worried about failing.

And one of the things I hope conviction narrative theory

will do is to help people to see

that if you want to succeed,

you've actually got to fail.

If you're always succeeding,

you can't be trying hard enough.

I really like this research funding

or resource allocation setting.

There's like these nested conviction narratives.

Like at the largest scale,

like we believe that funding basic research

or fundamental research or translational research

is going to be beneficial along these ways.

It's like we're convinced of that

and we're going to,

we're, we're holding on that.

And then at the program level,

we have a suspicion that

this or that aspect of this phenomena

is something that the program manager

is going to get behind

and that the resources are all aligned.

And then like in the review itself,

we might as well directly ask,

like to the proposer,

what are you convinced of?

And then not like just what are the risks

or uncertainty

or what makes you avoidance,
but like what are you convicted of
that makes you charge into that uncertainty
and super excited to carry out the research.
And we want to hear about the fuzziness
and uncertainty
because we don't even know simple things
until after the fact.
So we don't expect you to have the whole plan.
But at the same time,
it's like,
we do need to know
that your team and context
is convinced
so that you'll continue the approach
as uncertainty occurs
rather than upfront
so much consideration of the uncertainty
that you might never begin.
And then that sets up a narrative
where even small experimental perturbations,
which might be on the path
to transformative research,
that those have a negative valence
rather than like an almost motivational valence.
And that's just like,
that's one example
that could come into play
in funding allocation immediately.
Yeah, I think there's a lot of,
you know,
that's a very practical area
where it can be really thought about.
Do you have any final comments?
No, it's been nice talking to you.
And I do have,
I do find active inference
very interesting

and it kind of,
it's at a feeling level.

It feels right to me.

We may even use conviction narrative
in the active inference education setting.

That's like,
you could do a hundred PhDs
in thermodynamics
and you still wouldn't have
some bar of gold
waiting for you at the end.

That's like,
oh, now I understand equation 2.6.

So what is it then
that keeps us adding those layers of paint
in our learning journey?

I think it's an,
it's an excellent thought
and approach and sentiment.

And also if you provide the slides,
there are requests in the chat,
then I can post them alongside the video.

So thank you.

Thank you very much, Daniel.

Thank you, everyone.

See you.

Bye.

Bye.

Bye.

Thank you.